

Abhishek Singh

Singhabhishek1571@gmail.com — +91-9648531091 — LinkedIn — Portfolio — GitHub

Education

Central University of Chhattisgarh, Bilaspur

Dec 2022 – Apr 2026

Bachelor of Technology in Minor in AI

CGPA: 8.1

Experience

Speech Lab (IIT Madras) — Project Staff

- Reproduced and extended A-DMA for diffusion-based TTS using Conditional Flow Matching (CFM) and DiT Transformers with GPU-accelerated training.
- Implemented ConvNeXt text encoding with CTC and HuBERT-based SSL alignment for multimodal speech-text learning.
- Optimized diffusion TTS training using cosine similarity losses and PyTorch-based generative audio pipelines.

Indian Institute of Science (IISc), Bangalore (TANUH.ai)

Nov 2025 – May 2026

Machine Learning Research Intern

- Designed QPNN model for time-series forecasting on non-stationary cloud workloads, outperforming GRU/LSTM baselines.
- Built memory-efficient streaming training pipeline (less than 2 GB RAM) for large-scale Alibaba cluster datasets.
- Achieved 68–73% reduction in MSE across 1–20 min forecasting horizons through model optimization and tuning.
- Developed and evaluated end-to-end ML pipeline for preprocessing, training, and real-world deployment.

Indian Institute of Technology (IIT) Hyderabad

May 2025 – Jun 2025

AI/ML Research Intern

- Developed deep learning-based object detection models using PyTorch, OpenCV, and YOLOv5, achieving 92% accuracy under real-world conditions.
- Built an end-to-end computer vision pipeline integrating lane detection and multi-sensor fusion (LiDAR, GPS) for real-time decision-making.
- Improved model performance by 15% through hyperparameter tuning, optimization, and systematic evaluation.
- Validated model robustness across diverse lighting and environmental conditions to ensure reliable deployment.

Projects

Vertical.ai – Research-Grade RAG & Knowledge Intelligence Platform

Source Code

- Built RAG-based system for citation-grounded Q&A over documents, reducing manual reading effort by 60%.
- Implemented academic search and ingestion pipeline using OpenAlex and FAISS for citation-ranked retrieval.
- Designed interactive knowledge exploration with structured views and dynamic navigation for better understanding.
- Engineered scalable FastAPI backend with Groq LLMs and embeddings for low-latency AI responses.

Dodge AI – ERP Copilot (Order-to-Cash Intelligence Platform)

Source Code

- Built AI-powered ERP Copilot converting natural language queries into SQL for real-time data analysis.
- Developed FastAPI backend and React frontend with Cytoscape-based graph visualization for O2C workflows.
- Integrated AI agent for contextual insights, query understanding, and business-level explanations.
- Deployed scalable system on Render and Vercel ensuring low-latency and production reliability.

Smart Agriculture Monitoring System — AI, IoT, Precision Irrigation

Deployed Link

- Built an AI & IoT-based smart irrigation system for terrace farming using soil moisture, NPK, and climate data.
- Developed low-cost sensor micro-nodes with LoRa communication for real-time monitoring in hilly regions.
- Enabled precision irrigation and fertilizer recommendations using crop-stage and nutrient analysis.
- Reduced irrigation cost and water wastage through shared-tank architecture and adaptive scheduling.

Technical Skills

- **Programming Languages:** Python, SQL, Data Structures & Algorithms, Object-Oriented Programming
- **AI / ML:** Machine Learning, Deep Learning, Computer Vision, NLP, Time-Series Forecasting,
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, OpenCV, JAX, Selenium
- **Web & Backend Development:** React.js, React Native, Node.js, FastAPI, Flask, REST APIs
- **System Design:** HLD & LLD (Basics)
- **Tools:** Git, GitHub, Linux, Jupyter, Google Colab
- **AI Automation:** Langchain, Langgraph, LangSmith, Langflow, Crewai, Agno
- **Cloud & DevOps:** Microsoft Azure (Basics), Docker, CI/CD (Basics)
- **Databases:** MongoDB, MySQL, PostgreSQL

Hackathon & Achievements

- SIH 2025 – Winner
- BNi Shark Tank 25 – 2nd Winner
- Hack For Impact – Top 9 (IIT Delhi)
- Hacktivate 2025– Winner
- NIA 2024 – Winner
- TIF–Top 13 (IIT Madras)
- GGV Ideathon 2024– Winner
- Grant Recipient – CBDE
- HackVerse 2025 – Top 4